ADITYA UNIVERSITY

(Formerly known as Aditya Engineering College (A))

000403

PART-I

S.No: 421943

Programe	:	B.TECH
Hall Ticket No	:	22A91A0333
Name	:	OLETY ATCHUTA SRI SAI ARAVIND
Examination	:	III SEMESTER SUPPLEMENTARY (AR20(R))
Month-Year	:	Nov/Dec 2025
Branch	:	MECHANICAL ENGINEERING
Course Code	:	201BS3T12
Course Name	:	Integral Transforms And Applications Of PartialDifferential Equations
Date of Exam	:	18-11-2025

J. Pavan.

Signature of the Controller of Examination

O. Aravind 18-11-25

Signature of the Student with date

M. ... 18/11/25

Signature of the Invigilator with date

Serial No. Last Page Written	19

INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
   i. Writing their Hall Ticket No. and name in any part of the answer booklet.
   ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
   iii. Writing religious symbols.
   iv. Either seeking or providing any assistance to the fellow students in the examinations.
   v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
   vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
   vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**

SPACE FOR ROUGH WORK

Q.No.

Unit 1

1

a Given Problem,

find the fourier Series of $f(x) = \begin{cases} x & \text{if } 0 \leq x \leq 3 \\ 6-x & \text{if } 3 \leq x \leq 6 \end{cases}$

given that,

$$f(x) = \begin{cases} x & \text{if } 0 \leq x \leq 3 \\ 6-x & \text{if } 3 \leq x \leq 6 \end{cases}$$

formula using,

$$= \frac{a_0}{2} + \sum_{n=0}^{\infty} a_n \cos nx + b_n \sin nx$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} f(x)\, dx$$

$$a_n = \frac{1}{\pi} \int_0^{\pi} f(x) \cos nx\, dx$$

$$b_n = \frac{1}{\pi} \int_0^{\pi} f(x) \sin nx\, dx.$$

here $f(x) = f(x)$ so its a even funt so

$$= \frac{a_0}{2} + \sum_{n=0}^{\infty} a_n \cos nx\, dx.$$

Q.No.

$f(x) = x$ $\qquad$ $0, 3.$

$a_0 = \frac{2}{\pi}\int f(x)\,dx$

$= 2/\pi \int_0^3 f(x)\,dx$

$a_n \cdot 2/\pi \int_0^3 \cancel{3}1\,dx$ $\qquad$ $3dx = 0$

$a_n = 2/\pi \int_0 [1]_0^3$

$\boxed{a_0 = \frac{2\times3}{\pi}}$ $\qquad$ $\boxed{a_0 = 6/\pi}$

$a_n = 2/\pi \int_0^3 f(x)\cos nx\,dx$

$a_n = 2/\pi \int_0^3 \underset{u}{x} \underset{v}{\cos nx}\,dx$

$U \times V = UV_1 - U'V_2.$

$a_n = 2/\pi \int_0^3$

	U	V
u	x	$v = \cos nx.$
u'	1	$v_1 = \frac{-\sin nx}{n}$
u''	0	$v_2 = \frac{-\cos nx}{n}$

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Q.No.

1

b. Question given,

find the fourier series of $f(x) = x\cos x$ on interval of $(-\pi, \pi)$.

$f(x) = x\cos x$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{a} a_n \cos nx + b_n \sin nx.$$

$$a_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_0^{\pi} f(x) \cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_0^{\pi} f(x) \sin nx\,dx$$

here.

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} \cos nx\,dx.$$

diff of $\cos = -\sin x$

$\int \cos = \frac{\cos^2}{2}$

$\boxed{\sin\pi = 0}$

$$a_0 = \frac{1}{\pi}\left[\frac{\sin^2 nx}{2}\right]_0^{\pi}$$

$$a_0 = \frac{1}{\pi}[0].$$ $\boxed{a_0 = 0}$

Q.No.

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} \underset{U}{\cos x}\cdot\underset{V}{\cos nx}\,dx$$

$$UV = UV_1 - U'V_2 + \cancel{U''V_3}$$

U	V
$\cos x$	$\cos nx.$
$-\sin x$	$-\frac{\sin nx}{n}$
$-\cos x$	$-\frac{\cos nx}{n^2}$

$$\int_{-\pi}^{\pi}\left(\frac{\cos x \times \sin nx}{n} - \frac{\sin x - \cos nx}{n^2}\right)$$

$$\frac{1}{\pi}\ \cancel{\cos x}\int \frac{\sin nx \cos x}{n} - \frac{\sin x - \cos nx}{n^2}$$

$$\frac{1}{\pi}\left[\left[\frac{\sin x\cos nx}{n} - \frac{\sin x - \cos^2 nx}{n^2}\right]_{-\pi}^{\pi}\right.$$

$$\frac{1}{\pi}\left[\frac{\sin \pi \cos \pi}{n} - \frac{\sin \pi\ \cos^2 n}{n^2}\right]$$

$$= \frac{1}{\pi}\left[\frac{1}{n^2}\right]$$

$$\boxed{a_n = \frac{1}{\pi n^2}}$$

Q.No.

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx \, dx.$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \underset{u}{\cos x} \; \underset{v}{\sin nx \, dx}$$

u v

$\cos x$ $\sin nx \, dx$

$-\sin x$ $\frac{\cos nx}{n}$

$-\frac{\sin nx}{n}$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \frac{\cos x \cos nx}{n} - \frac{\sin nx \times \sin}{n}$$

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \frac{\cos (nx)}{n} - \frac{\sin (n^2 x)}{n}$$

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \cos x - \sin (nx)$$

$$= \frac{1}{\pi} \left[\cos x - \sin (nx) \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\cos \pi n \; \sin n\pi \right]_0^{\pi}$$ $\sin \pi = 0$

$\cos \pi = 1.$

$$= \frac{1}{\pi} \cos \pi n$$

$$\boxed{b_n = \cos \pi n / \pi}$$

Q.No.

after Solve all a_0, a_n, b_n the we

can keep them on formula

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx.$$

Sub all Value in here.

$$= \frac{0}{2} + \sum_{n=1}^{\infty} \frac{1}{\pi n^2} \cos nx + \frac{\cos \pi n}{\pi} (\sin nx)$$

$$= \sum_{n=1}^{\infty} \frac{\cos n}{\pi n^2} + \frac{\cos \pi n \times \sin nx}{\pi}$$

Q.No.

Theorem used

Convolution theorem.

$$L^{-1}\left\{\frac{s^2}{(s^2+4)(s^2+9)}\right\}$$

$$\bar{f}(\cancel{s}) \cancel{*} \bar{g}(s) = \bar{f}(g \cancel{*} F)...$$

$$f(g)\, f(t) = \int_0^{t} e^{...} + e^{...}$$

$$L^{-1}\left\{\frac{1}{s^2+4}\right\} = F(s) = e^{t}$$

$$L^{-1}\left\{\frac{s^2}{s^2+9}\right\} = g(s) = e^{bt}$$

Q.No.

$$\cancel{f(t)} = \int_0^t e^{au} + e^{bt} \, u g(u)$$

$$= \int_0^t e^{u} + e^{bt(u)} \, u$$

$$= e^{u} + e^{bu} - e^{bu}$$

$$= e^{u} \int e^{u} (e^{b} - e^{t})$$

add ay only

$$\cancel{\ln} \frac{1}{u} = \cancel{\log} e^{nu} \int \ln e^{b} - e^{t}$$

$$= 1/x^2 + 2bh + e^{nu} - n$$

Q.No.

7 given questions

Find $L^{-1}\left\{\frac{2s-3}{s^2+4s+13}\right\}$

here we are going to exam

elimd the unt 2 part.

let

$L^{-1}\left\{\frac{1}{s^2+4s+13}\right\}$

$L^{-1}\left\{\frac{2s-3}{s^2-4s+3}\right\}$.

Q.No.

9. $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$

$u(x,0) = 6e^{-3x}$

$\partial u = X'T$

$\partial x = XT'$

$X'T = \frac{2\partial u}{\partial t}$

$XT' = 6e^{-3x}$

usin method of Seperation Seprat the an,

$\log\frac{1}{x} = \frac{2\partial u}{\partial t}$

here meth of XX' & TT' are the varules of thens.

the

$\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$

the $\frac{2\partial u}{\partial t}$

the nnt $XX' = \frac{2\partial u}{\partial t}$.

Q.No.

Q.No.

Unit - II

3a. $\int_0^\infty \frac{\cos\lambda x}{1+\lambda^2} = \pi/_2 e^{-x} \quad x \geq 0$

The

$$f(x) = \int_0^\infty \frac{\cos\lambda x}{1+\lambda}$$

$$\Rightarrow a_0 \sum_{n=1}^{\infty} a_n \int e^{nx} f(x).$$

$$a_0 = \frac{2}{\pi} \int \frac{\cos\lambda x}{1+\lambda^2}.$$

$$1/_2 = \frac{\cos\lambda x + \sin\lambda x \Rightarrow}{1+\lambda^2}$$

$$\cos\lambda x + \cos\lambda x + \sin$$

$$\frac{2(\cos\lambda x) + \sin}{1+\cancel{x}^2}$$

$$\int_0^\pi 2\cos x + \sin$$

Q.No.

$$\left[2x\cos x\pi + \sin x \pi\right]_0^{4}$$

$2x \cos x +$ $\quad\quad$ $\therefore \cos 2\pi = 1 \times \cos$

$$= \frac{2\sin + 2\sin}{2} + \sin$$

The $\sum \cos x \ldots$

$$= \cos \sum \cos x \frac{\cos x}{1+\lambda^2}$$

$$= \frac{\cos^2 x}{1+\lambda^2}$$

$$= \frac{\cos^2 x + \sin^2 x}{1+\lambda^2}$$

$$= \frac{1}{1+\lambda^2} + 2$$

$$= \frac{2(1+\lambda^2)+1}{1+\lambda^2}$$

$$\frac{2+2\lambda^2+1}{1+\lambda^2}$$

Q.No.

$$= \frac{2+1+2\lambda^2}{1+\lambda^2}$$

$$1+\lambda^2 = 3+2\lambda^2$$

$$\cancel{3}\ 1+\lambda^2 - 3 = 2\lambda^2$$

$$\lambda^2 = 3+2\lambda^2 - 1$$

$$\lambda^2 = \cancel{3}\ 2\lambda^2 - 1 + 3$$

$$\lambda^2 = 2\lambda^2\ 2$$

$$\cancel{\lambda^2 = 4\lambda^3}$$

$$\lambda^{3} = \lambda^2 4$$

$$\boxed{\lambda = 4}$$

then $\int_0^a \frac{\cos 4x}{1+4^2}$

$= \int_0^a \frac{\cos 4x}{17}$

Q.No.

3

b)

The

an and an $f(x) = \begin{cases} 1 & \text{if } 0 \le x < 2 \\ 0 & \text{if } x \ge 2 \end{cases}$

The

$$\frac{a_0}{2} + \sum_{n}^{\infty} a_n \cos nx + b_n \sin nx.$$

$a_0 = \frac{1}{4\pi} \int_{-4}^{4} f(x) dx$

$a_n = \frac{1}{4\pi} \int_{-4}^{4} f(x) \cos nx \, dx$

$b_n = \frac{1}{4\pi} \int_{-4}^{4} f(x) \sin nx \, dx.$

Q.No.

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SEMI-LOG PAPER (5 CYCLES X 1/10")